Commentary on Tasks 1- 4 in Coursework 1.2

This analysis uses the USArrests dataset in R, which consists of arrests per 100,000 residents for murder, assault and rape in each of the 50 states in 1973. This analysis examines the quality of the data, the descriptive statistics, how it is distributed, the correlation between variables and the limitations of the dataset while interpreting the data.

In task 1, we were required to check for the validity and quality of values in the dataset.

#validate_data(USArrests)

This function was created to test if there were duplicates, non-numeric analysis columns, negative values, and values outside plausible ranges. This function returned FALSE results for these factors. However, passing this function's checks does not guarantee accuracy, as this validation is limited to the conditions in which the function was programmed.

Statistics	Murder	Assault	UrbanPop	Rape
Minimum	0.8	45	32	7.3
Maximum	14.4	337	91	46
Mean	7.788	170	65.54	21.232
Median	7.25	159	66	20.1
Standard Deviation	4.356	83.338	14.475	9.366
Interquartile range	7.175	140	23.25	11.1

In task 2, Using,

calculate_stats()

We calculated descriptive statistics to examine the center, spread, and possible skewness of the four variables. Assault had the highest variability, with an interquartile range of 140 and a standard deviation of 83.34. The means of Murder, Assault, and Rape were higher than their medians, indicating possible right skewness.

Michael Brenndoerfer (2025) says histograms display the distribution of a continuous variable by dividing the range of values into intervals and counting observations in each interval. Box plots provide a concise visual summary of a distribution's key statistical properties. Murder and Rape appeared right-skewed, Assault appeared bimodal, and UrbanPop looked symmetric, with the highest frequency in the 60–70% interval. The boxplots showed variation in Assault arrest rates and two potential outliers for Rape, representing Alaska (44.5) and Nevada (46.0). These instances may have contributed to the right skewness of the Rape distribution, but it it does not make them the determinant for errors.

Frequency labels were included in the histogram charts, and the quartile values were included as labels on the boxplots for easier reading. A limitation is that the patterns on the histogram depend on the selected bin widths, and potential outliers on boxplots do not automatically determine data error. The visualizations generally help to examine patterns and differences between the 50 states, but the results should be analyzed in view of the dataset's limitations.

We calculated and plotted Pearson and Spearman correlation coefficients to check for the correlation between Murder, Assault, Rape, and UrbanPop. J.C.F. De Winter et al. (2016) say that the Pearson correlation coefficient measures the linear relationship between two continuous random variables, while Spearman correlation measures the monotonic relationship. Correlation heatmaps were used to compare the coefficients, while ggpairs() combined the scatterplots, correlation values, and distributions for more visual results.

These two methods showed Murder and Assault to have the strongest correlation coefficient (Pearson = 0.802; Spearman = 0.817), while Murder and UrbanPop had the weakest correlation coefficient(Pearson = 0.070; Spearman = 0.107). The "Murder-Rape correlation" was stronger using Spearman (0.679) than Pearson (0.564); this suggests that the relationship may not be completely linear. But this contrast alone should not define the exact shape of the relationship.

Both correlation methods provide insight into relationships between variables, but neither proves that one crime happens because of the other. The dataset records arrest

rates rather than actual crimes committed, which may be influenced by reporting rates, police enforcement, and cultural or racial biases. Additionally, since the data is from 1973, it may not reflect current crime patterns.

The dataset includes only four variables and excludes social factors such as education, security, economic inequality, and demographics. Therefore, the correlations cannot explain why differences occur or whether one factor causes another.

These findings may help identify patterns for further investigation but should not be used alone to guide crime prevention, prediction, or police resource allocation. More recent data, additional variables, and state-level research comparing crimes committed with arrests made are needed for more reliable decisions.

In conclusion, the analysis revealed differences in arrest rates across 50 US states. Assault showed the highest variability, while Murder and UrbanPop had the weakest positive correlation and Murder and Assault had the strongest. However, the dataset’s age, limited variables, and reliance on arrest rates rather than actual crime figures mean the results should be interpreted cautiously.

References

De Winter, J.C.F., Gosling, S. D., & Potter, J. (2016). “COMPARING THE PEARSON AND SPEARMAN CORRE4LATION COEFFICIENTS ACROSS DISTRIBUTIONS AND SAMPLE SIZES: A TUTORIAL USING SIMULATIONS AND EMPIRICAL DATA.” *Psychological Methods*, 21(3), 273-290. Available at https://doi.org/10.1037/met0000079 [https://arxiv.org/pdf/2408.15979]. [Accessed October 23, 2026]

Michael Brenndoerfer (2025) 'Data Visualization: Histograms, Box Plots and Scatter Plots'. Available at: Data Visualization: Histograms, Box Plots and Scatter Plots - Interactive (Accessed: October 26, 2026).